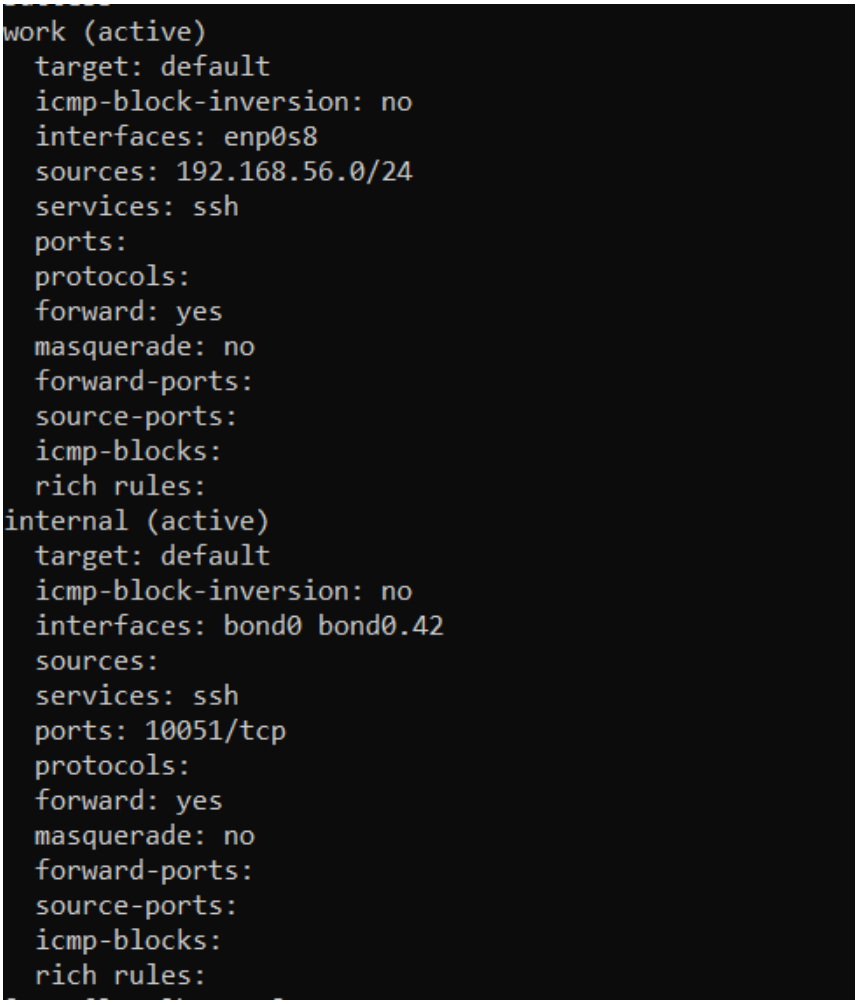
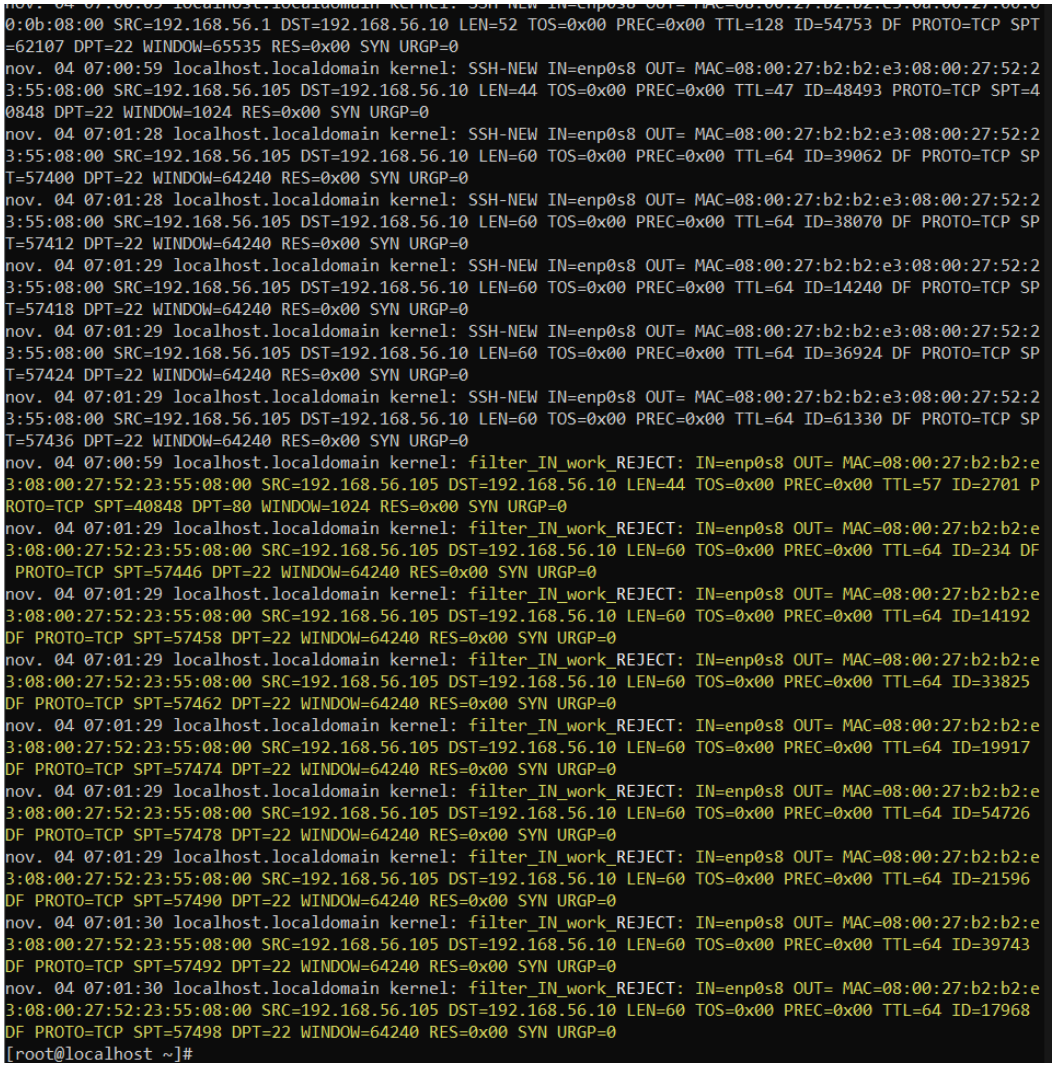


1. Zones firewalld : SSH autorisé depuis le réseau 192.168.56.0/24, port 10051 ouvert en zone internal.

```
[root@srv-edge ~]# nmap -Pn -p 22,80,10051 192.168.56.10 | tee /tmp/nmap_ava
nt_ou_apres.txt
nc -zv 192.168.56.10 22
nc -zv -w 2 192.168.56.10 10051 ; echo "rc attendu != 0 : $?"
for i in $(seq 1 12); do nc -zv -w1 192.168.56.10 22 ; done
tee: /tmp/nmap_avant_ou_apres.txt: Permission non accordée
Starting Nmap 7.92 ( https://nmap.org ) at 2026-11-04 12:39 CET
Nmap done: 1 IP address (0 hosts up) scanned in 1.55 seconds
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
rc attendu != 0 : 1
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )

Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )
```

2. Test des ports 22, 80 et 10051 depuis srv-edge avec nmap et nc.



3. Journal du noyau : les tentatives SSH-NEW et les rejets du pare-feu sont enregistrés.

```
[root@srv-edge ~]# ssh sysops@192.168.56.10 'hostname'
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
localhost.localdomain
```

4. Bannière d'accueil affichée lors d'une connexion SSH au compte sysops.

```
[root@srv-edge ~]# ssh root@192.168.56.10 'id' ; echo "rc: $?"
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
root@192.168.56.10: Permission denied (publickey).
rc: 255
```

5. Connexion root refusée : seule l'authentification par clé est acceptée.



6. Tests après durcissement : ancien port 22 fermé, accès au port 2242 et hors ssh-admins refusés.

```
.ssh/authorized_keys:2
nov. 04 08:39:11 localhost.localdomain sshd-session[26210]: Accepted publicke
y for sysops from 192.168.56.105 port 45970 ssh2: ED25519 SHA256:sHNc3NBvWlw2
GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0
nov. 04 08:47:14 localhost.localdomain sshd-session[26680]: Accepted key ED25
519 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/
.ssh/authorized_keys:2
nov. 04 08:47:14 localhost.localdomain sshd-session[26680]: Accepted key ED25
519 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/
.ssh/authorized_keys:2
nov. 04 08:47:14 localhost.localdomain sshd-session[26680]: Accepted publicke
y for sysops from 192.168.56.105 port 40670 ssh2: ED25519 SHA256:sHNc3NBvWlw2
GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0
nov. 04 08:51:47 localhost.localdomain sshd-session[27303]: Accepted key ED25
519 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/
.ssh/authorized_keys:2
nov. 04 08:51:47 localhost.localdomain sshd-session[27303]: Accepted key ED25
519 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/
.ssh/authorized_keys:2
nov. 04 08:51:47 localhost.localdomain sshd-session[27303]: Accepted publicke
y for sysops from 192.168.56.105 port 44096 ssh2: ED25519 SHA256:sHNc3NBvWlw2
GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0
-- Boot beef2d5c75f347beb5c21a634ce29cd0 --
nov. 04 14:36:06 localhost.localdomain sshd-session[2184]: Accepted key ED255
19 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/.
ssh/authorized_keys:2
nov. 04 14:36:06 localhost.localdomain sshd-session[2184]: Accepted key ED255
19 SHA256:sHNc3NBvWlw2GoBcj1w8eVWZCgztuIlYsf/zzDmBIU0 found at /home/sysops/.
ssh/authorized_keys:2
nov. 04 14:36:06 localhost.localdomain sshd-session[2184]: Accepted publickey
 for sysops from 192.168.56.105 port 39246 ssh2: ED25519 SHA256:sHNc3NBvWlw2G
oBcj1w8eVWZCgztuIlYsf/zzDmBIU0
[root@localhost ~]# journalctl --verify | tail -5
journalctl --disk-usage
PASS: /var/log/journal/ed257528b4be4aafadfe7d3a6187fd75/user-1000.journal
PASS: /var/log/journal/ed257528b4be4aafadfe7d3a6187fd75/system@00065f64481b77
bc-46aa0c28ece55f48.journal~
PASS: /var/log/journal/ed257528b4be4aafadfe7d3a6187fd75/system.journal
PASS: /var/log/journal/ed257528b4be4aafadfe7d3a6187fd75/user-1001.journal
Archived and active journals take up 32.0M in the file system.
```

7. Connexions réussies par clé enregistrées dans le journal systemd.

```
[root@srv-edge ~]# for i in $(seq 1 10); do
  ssh -p 2242 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout=3 \
    pirate@192.168.56.10 true 2>/dev/null
done
```

8. Dix tentatives de connexion de l'utilisateur pirate, toutes enregistrées.

```
1 timers listed.
Pass --all to see loaded but inactive timers, too.

[root@localhost ~]# journalctl -u sshd --since '-5min' -g 'Invalid user' --no
-pager | tail
systemctl start helios-watch-auth.service
journalctl -t helios-watch-auth -n 10 --no-pager
firewall-cmd --ipset=helios-blacklist --get-entries
nov. 04 14:42:57 localhost.localdomain sshd-session[2856]: Invalid user pirat
e from 192.168.56.105 port 51212
nov. 04 14:42:57 localhost.localdomain sshd-session[2858]: Invalid user pirat
e from 192.168.56.105 port 51224
nov. 04 14:42:57 localhost.localdomain sshd-session[2860]: Invalid user pirat
e from 192.168.56.105 port 51236
nov. 04 14:42:57 localhost.localdomain sshd-session[2862]: Invalid user pirat
e from 192.168.56.105 port 51240
nov. 04 14:42:57 localhost.localdomain sshd-session[2864]: Invalid user pirat
e from 192.168.56.105 port 51252
nov. 04 14:42:58 localhost.localdomain sshd-session[2866]: Invalid user pirat
e from 192.168.56.105 port 51254
nov. 04 14:42:58 localhost.localdomain sshd-session[2868]: Invalid user pirat
e from 192.168.56.105 port 51258
nov. 04 14:42:58 localhost.localdomain sshd-session[2870]: Invalid user pirat
e from 192.168.56.105 port 51262
nov. 04 14:42:58 localhost.localdomain sshd-session[2872]: Invalid user pirat
e from 192.168.56.105 port 51274
nov. 04 14:42:58 localhost.localdomain sshd-session[2874]: Invalid user pirat
e from 192.168.56.105 port 51284
Job for helios-watch-auth.service failed because the control process exited w
ith error code.
See "systemctl status helios-watch-auth.service" and "journalctl -xeu helios-
watch-auth.service" for details.
-- No entries --
```

9. Journal sshd des utilisateurs invalides, puis démarrage du service de blocage.

```
[root@srv-edge ~]# for i in $(seq 1 10); do
  ssh -p 2242 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout=3 \
    pirate@192.168.56.10 true 2>/dev/null
done
[root@srv-edge ~]# nc -zv -w2 192.168.56.10 2242 ; echo "rc: $?"
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
rc: 1
```

10. Après les tentatives, le port 2242 ne répond plus à l'adresse bloquée.

```
[root@localhost ~]# echo "=== Liste de blocage ==="
firewall-cmd --ipset=helios-blacklist --get-entries
echo "=== Timer ==="
systemctl list-timers helios-watch-auth.timer --no-pager
echo "=== Journal du script ==="
journalctl -t helios-watch-auth -n 5 --no-pager
=== Liste de blocage ===

=== Timer ===
NEXT          LEFT          LAST          PASSED  UNIT          ACTIVATES
Wed 2026-11-… 1min 11s left Wed 2026-11-… 45s ago helios-watch… helios-watch…

1 timers listed.
Pass --all to see loaded but inactive timers, too.
=== Journal du script ===
nov. 04 14:48:04 localhost.localdomain helios-watch-auth[3282]: adresse 192.1
68.56.105 bloquee (10 echecs en 10 min)
nov. 04 14:48:04 localhost.localdomain helios-watch-auth[3283]: passage termi
ne
nov. 04 14:50:03 localhost.localdomain helios-watch-auth[3405]: adresse 192.1
68.56.105 bloquee (10 echecs en 10 min)
nov. 04 14:50:03 localhost.localdomain helios-watch-auth[3406]: passage termi
ne
nov. 04 14:52:03 localhost.localdomain helios-watch-auth[3520]: passage termi
ne
```

11. Liste de blocage helios-blacklist, timer du script et journal du blocage.